CMOS Inverter Layout Design

Physical Verification and Layout Parasitic Extraction

Technology Node: SAED 32/28nm Tools: Synopsys Custom Compiler, IC Validator

Overview

This document records the physical design of a CMOS inverter implemented using the SAED 32/28nm PDK in Synopsys Custom Compiler. The work includes transistor-level layout creation, inverter layout integration, physical verification, and layout parasitic extraction.

The layout flow begins with the independent creation of PMOS and NMOS device layouts. These devices are then instantiated and integrated to form the inverter cell. After routing the required connections for input, output, power, and ground, the layout is verified using IC Validator through DRC and LVS. Finally, the layout is prepared for parasitic extraction to support post-layout simulation.

Document Flow

1. Design specifications
2. PMOS transistor layout
3. NMOS transistor layout
4. Inverter layout integration
5. Physical verification (DRC and LVS)
6. Layout parasitic extraction
7. Design flow summary

1 Design Specifications

The inverter was designed using a PMOS width of $1\,\mu m$ and an NMOS width of $0.5\,\mu m$. Both devices use a gate length of $30\,nm$. The PMOS width is chosen to be approximately twice the NMOS width to compensate for the lower hole mobility in PMOS devices and to achieve more balanced rise and fall transitions at the output.

Parameter	Value	Remarks
PMOS Width	$1.0\,\mu m$	Chosen as 2x NMOS width
NMOS Width	$0.5\,\mu m$	Baseline pull-down sizing
Gate Length	$30\,nm$	Minimum drawn gate length
Technology Node	SAED 32/28nm	Academic PDK
Verification Tool	IC Validator	DRC and LVS
Extraction Flow	LPE / Star-RCXT	RC extraction for post-layout simulation

Why PMOS is Wider

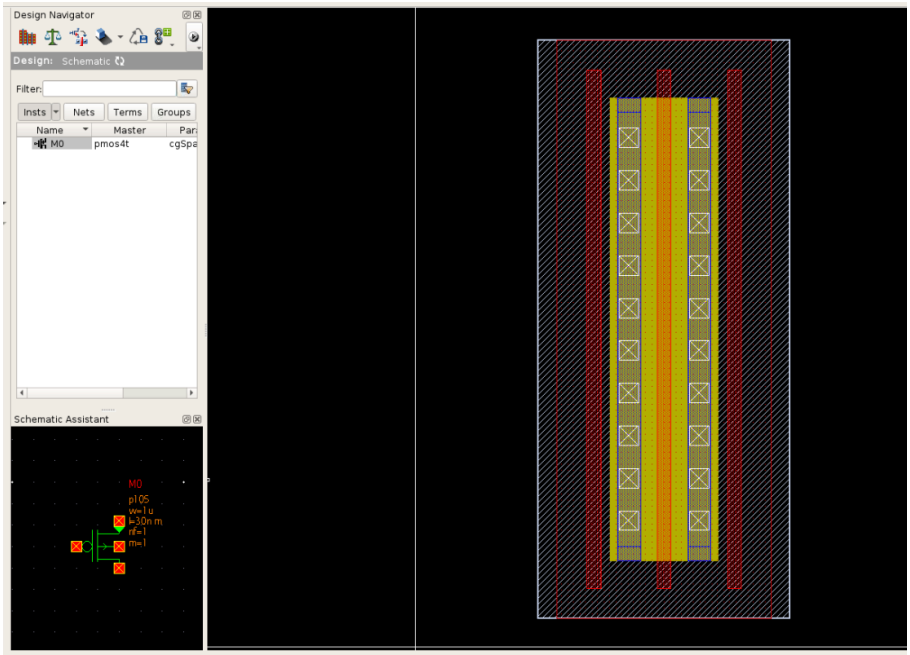
The PMOS transistor is made wider than the NMOS because hole mobility is lower than electron mobility. If both transistors were drawn with the same width, the PMOS would provide less current and the output rise transition would be slower than the fall transition. A wider PMOS helps balance the effective drive strength of the pull-up and pull-down paths.

2 Transistor Layout Creation

Each transistor was first created as an individual layout cell. This made it easier to verify the PMOS and NMOS structures independently before integrating them into the final inverter layout.

2.1 PMOS Layout

The PMOS transistor layout was created first using a width of $1\,\mu m$ and a gate length of $30\,nm$. The gate was defined using a poly strip drawn vertically through the active region.



PMOS Device Layout

Gate Formation

A poly strip was used to define the gate. Since the design uses a gate length of 30 nm, the poly dimension defining the channel length was drawn accordingly. Wherever the poly crosses the active region, the MOS channel is formed.

Why Poly Defines Channel Length

In MOS layout, the gate is formed where the poly crosses the active region. The channel exists directly under the gate oxide in this overlap region. Therefore, the width of the poly in the current-flow direction determines the drawn channel length.

Dummy Poly

Dummy poly was added on both sides of the functional gate. These dummy structures do not carry electrical signals but are included to improve fabrication uniformity and satisfy density-related design requirements.

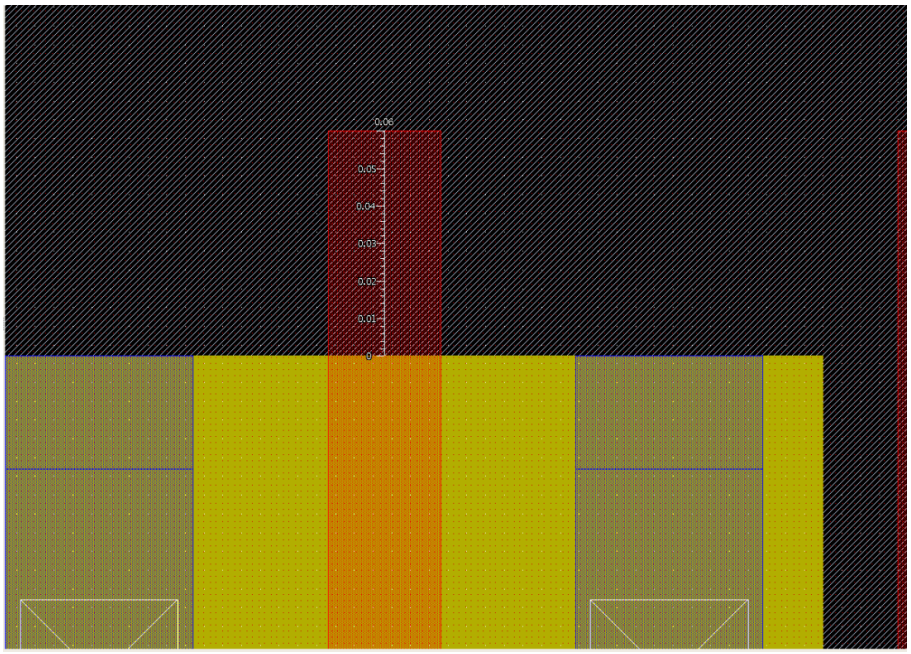
Poly Spacing

The poly-to-poly spacing was maintained according to DRC. In this layout, the minimum poly-to-poly spacing used was 0.152 μm, verified using IC Validator.

Active Region

The active region was drawn with a height of 1 μm for the PMOS width. The side spacing from active to poly was maintained as 0.02 μm, and the top extension beyond the active

was maintained as $0.06\,\mu m$.



Poly Extension Beyond Active Region

Why Poly Extends Beyond Active

The poly gate extends beyond the active region to ensure proper gate formation even in the presence of lithography and alignment tolerances. If the poly ended exactly at the active boundary, slight misalignment during fabrication could result in incomplete gate overlap and unreliable transistor behavior. The extension also satisfies the enclosure rule defined by the PDK.

Implant Layer

A P-implant layer was drawn over the PMOS active region. The poly-to-implant spacing on each side was maintained as $0.065\,\mu m$.

PMOS Implant and Poly-to-Implant Spacing

Purpose of Implant Layers

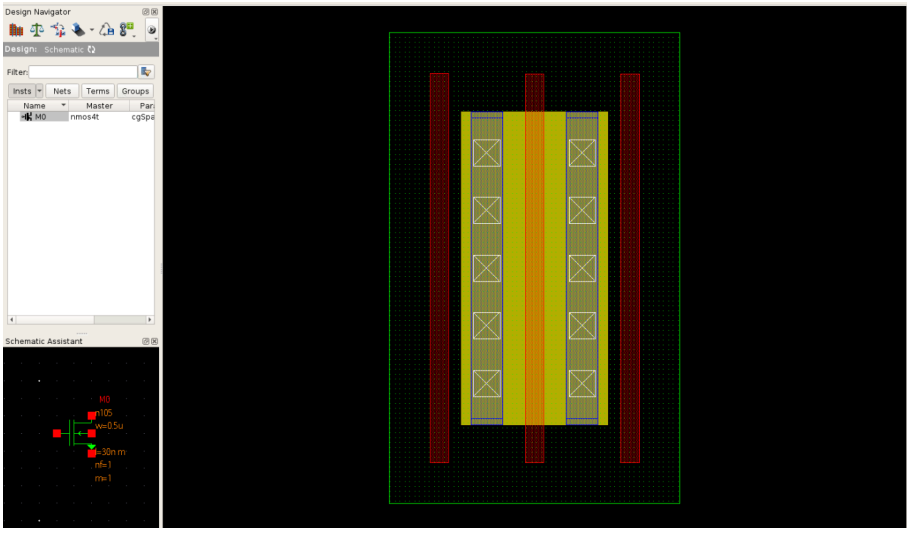
Implant layers define the required doping in the source and drain regions. For PMOS, the implant mask ensures that the active regions form p-type source and drain junctions with the required electrical characteristics.

Contacts

Contacts were added on the source and drain diffusion regions to connect the device to metal routing. Proper enclosure and spacing rules were observed while placing the contact array.

2.2 NMOS Layout

The NMOS transistor was created using the same sequence as the PMOS layout, with a width of $0.5\,\mu m$ and the same gate length of $30\,nm$. The main structural difference is that the NMOS does not require an N-well.

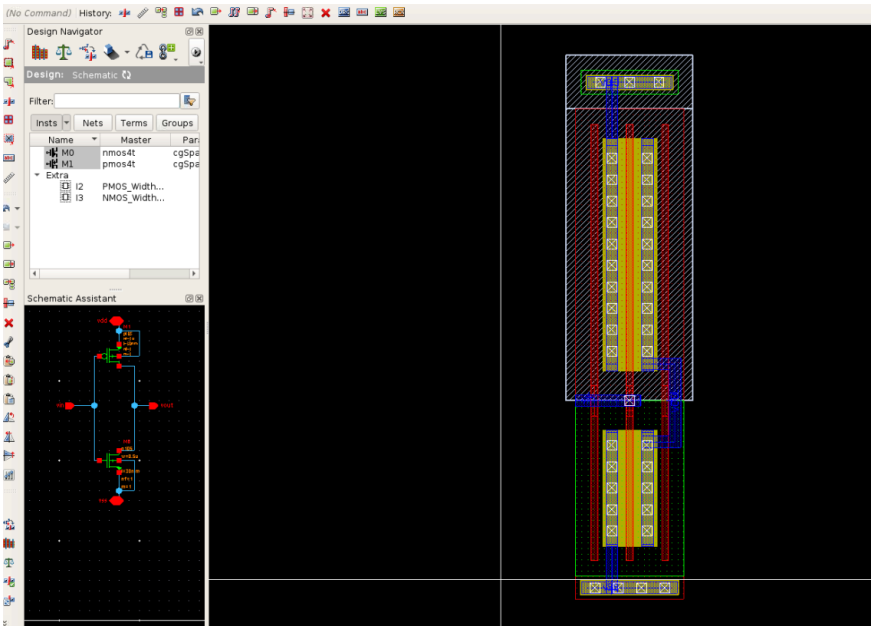


NMOS Device Layout

The active height for the NMOS was set to 0.5 μm, consistent with the chosen transistor width. The same poly spacing, dummy poly placement, and contact rules were followed. An N-implant was used for the NMOS source and drain regions.

3 Inverter Layout Integration

After completing the PMOS and NMOS layouts individually, both devices were instantiated into the inverter layout cell. The PMOS was placed in the upper region inside the N-well, and the NMOS was placed below it in the substrate region.



CMOS Inverter Layout After Device Integration and Routing

Device Placement

The PMOS was placed at the top of the layout to connect easily to the VDD rail, while the NMOS was placed at the bottom to connect to the VSS rail. This vertical arrangement provides a compact and standard inverter layout structure.

Routing and Connections

The gates of the PMOS and NMOS were connected together to form the input node *vin*. The drains of both devices were connected together to form the output node *vout*. The PMOS source was connected to VDD, and the NMOS source was connected to VSS. Routing was performed using metal layers after contacts were placed on the required source, drain, and gate regions.

Well and Substrate Connections

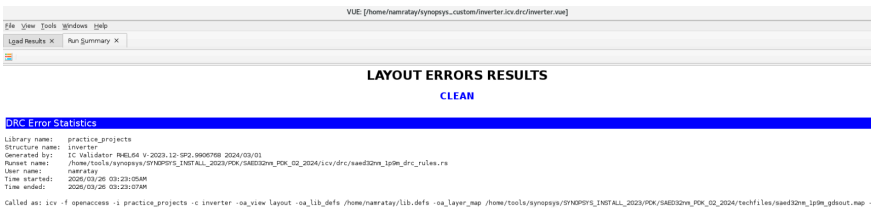
The PMOS device was enclosed within an N-well. A well tie was connected to VDD to fix the PMOS body potential. Similarly, a substrate tap was connected to VSS to maintain the NMOS body at ground potential and reduce latch-up risk.

4 Physical Verification

After completing the layout, physical verification was carried out using IC Validator.

4.1 Design Rule Check (DRC)

DRC was run using the DRC runset file provided by the SAED 32/28nm PDK. This check verified all geometry-related rules including poly spacing, active spacing, implant spacing, contact enclosure, metal width, and well enclosure.

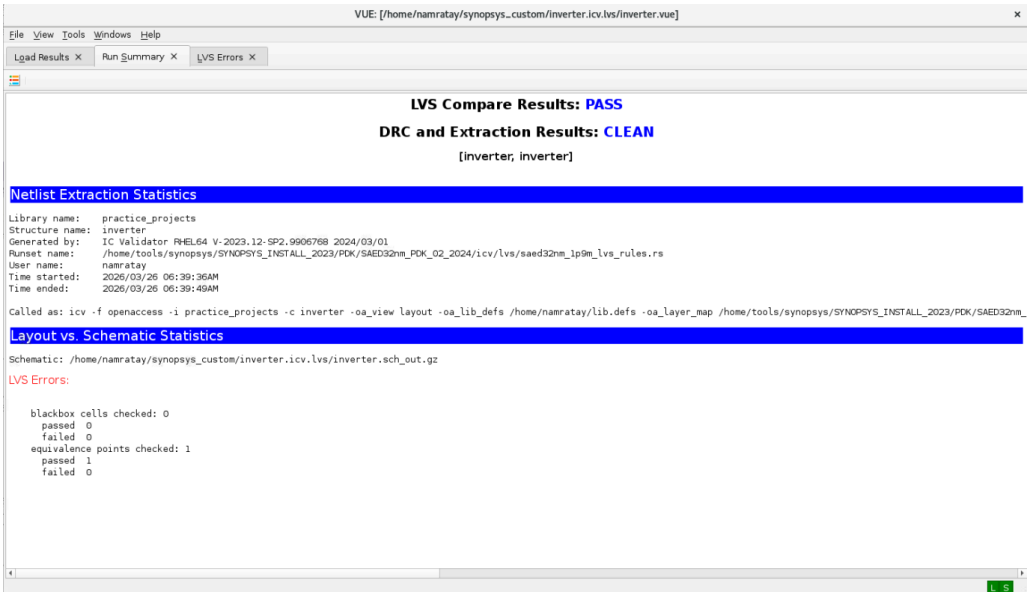


Layout View After DRC-Clean Verification

The layout was corrected iteratively until all DRC violations were removed and a clean result was obtained.

4.2 Layout Versus Schematic (LVS)

LVS was performed using IC Validator to compare the extracted layout netlist with the original schematic netlist. This step ensures that the devices, their dimensions, and interconnections in the layout match exactly with the intended schematic design.



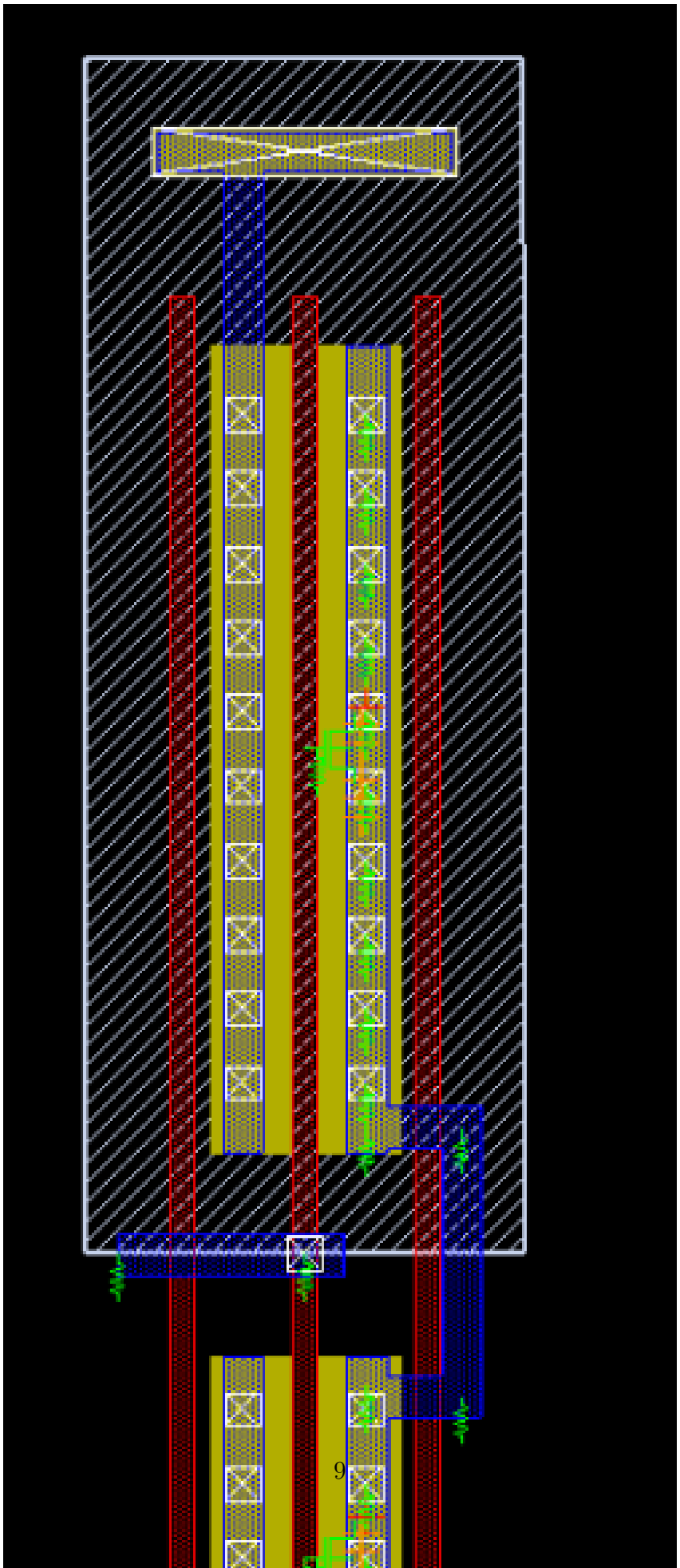
LVS Comparison Result Showing PASS and Clean Extraction

The LVS result shows a successful match between layout and schematic, indicating correct device connectivity and implementation. No mismatches were observed, confirming that the inverter layout is electrically equivalent to the schematic design.

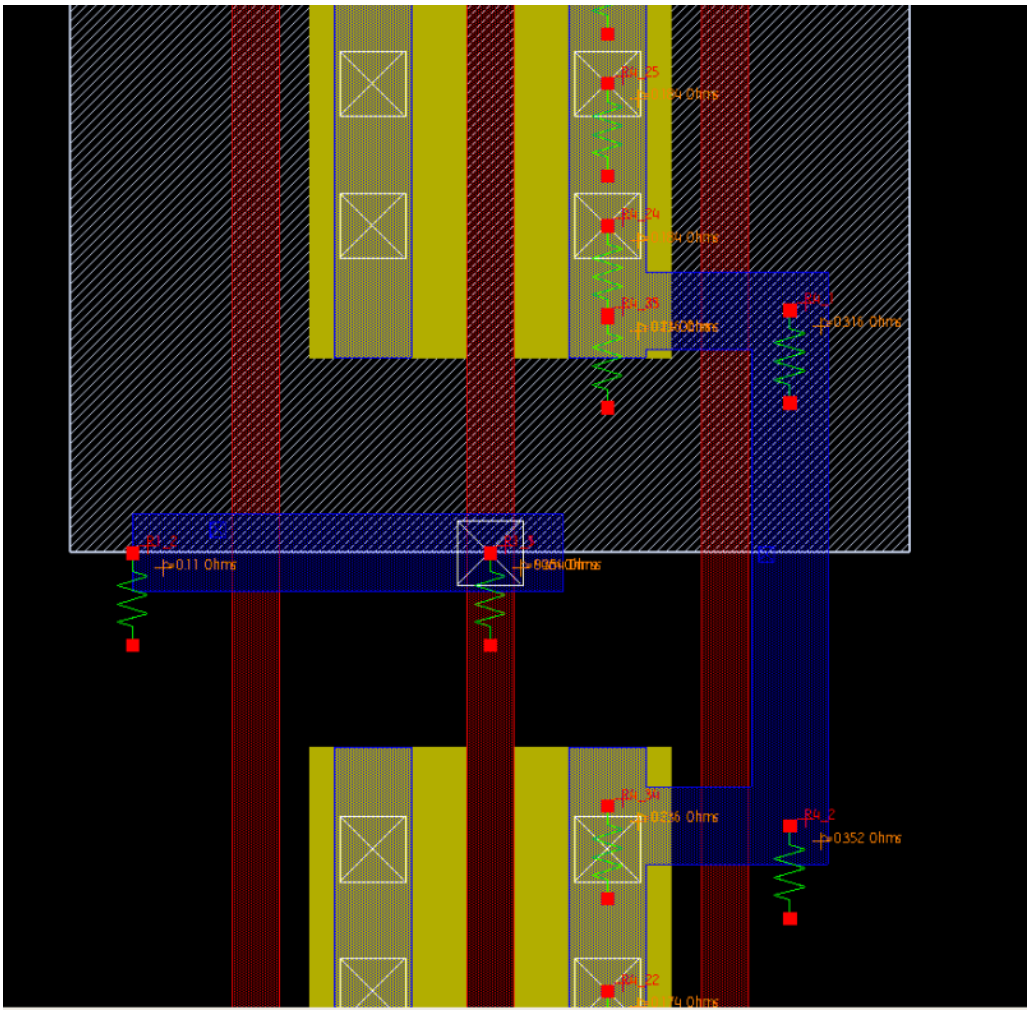
5 Layout Parasitic Extraction (LPE)

After successful DRC and LVS verification, layout parasitic extraction was performed to capture the non-ideal effects introduced by the physical layout. This step extracts parasitic resistance and capacitance from interconnects, contacts, and diffusion regions.

Extracted Layout View



Zoomed View of Parasitics



Zoomed View Showing Extracted Parasitic Resistances

Parasitic Resistance

Parasitic resistance is introduced due to:

- Metal interconnect paths
- Contacts and vias
- Diffusion regions (source/drain resistance)

In the zoomed view, the extracted resistances are visible along routing paths. These resistances increase with wire length and affect signal propagation delay.

Parasitic Capacitance

Parasitic capacitance arises due to:

- Overlap between different metal layers

- Gate-to-diffusion capacitance
- Coupling between nearby interconnects

These capacitances store charge and slow down transitions, especially during switching events.

Impact on Circuit Performance

The combined effect of parasitic resistance and capacitance results in RC delay, which affects:

- Propagation delay
- Rise and fall times
- Overall switching speed

Including these parasitic elements enables more accurate post-layout simulation compared to ideal pre-layout simulations.

6 Design Flow Summary

The CMOS inverter was designed following a structured workflow starting from schematic design and progressing towards physical implementation and verification.

Step 1: Schematic Design and Pre-Layout Simulation

The inverter schematic was created using PMOS and NMOS transistors with appropriate sizing.

Pre-layout simulation was performed in PrimeWave to verify functional correctness and observe switching behavior under ideal conditions.

Step 2: Transistor-Level Layout Creation

Individual layouts for PMOS and NMOS devices were created using the SAED 32/28nm PDK. Design rules such as poly spacing, active enclosure, and implant spacing were carefully followed and verified using DRC.

Step 3: Inverter Layout Integration

The PMOS and NMOS layouts were instantiated into a single layout cell. Connections were made as follows:

- Gates connected to form input (vin)

- Drains connected to form output (vout)
- PMOS source connected to VDD
- NMOS source connected to VSS

Routing was completed using metal layers while ensuring compliance with layout rules.

Step 4: Physical Verification

Physical verification was carried out using IC Validator:

- DRC was performed using the DRC runset file to ensure all geometric rules were satisfied
- LVS was performed using the LVS runset file to confirm that the layout matches the schematic

The layout was verified to be DRC clean and LVS matched.

Step 5: Layout Parasitic Extraction

After verification, parasitic extraction was performed using the layout netlist and RC extraction runset. This step captures parasitic resistance and capacitance from the physical layout to enable accurate post-layout analysis.

Overall flow followed:

$$\text{Schematic} \rightarrow \text{Pre-layout Simulation} \rightarrow \text{Layout Creation} \rightarrow \text{DRC} \rightarrow \text{LVS} \rightarrow \text{LPE}$$